MORPHISMS OF LIMITS LIFT TO MORPHISMS OF SYSTEMS OVER GOOD DIAGRAMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5359-5401, following the statement at 5323-5357.

Editorial note. No omissions; the cartesian identity for E_i is displayed as an equation chain.

Allowed standard prerequisites (editorial).

- Limit descriptions from good diagrams. lemma-limit-from-good-diagram (same chapter, cited at ~5344-5346). X = lim X_i and Y = lim Y_i.
- Characterization of locally finite presentation spaces. proposition-characterize-locally-finite-presentation (same chapter, card L26289, cited at ~5361-5363). Existence of h : X_{i_0} -> V.
- Fibre product criterion for relative limit preservation. lemma-better-characterize-relative-limit-preserving (same chapter, card L26304, cited at ~5361-5363). The improved form of the proposition applied to V -> B_{i_2}.
- Factorization through scheme theoretic images. Morphisms of Spaces, Lemma spaces-morphisms-lemma-factor-factor (cross-chapter, cited at ~5376-5381). X_i -> B_i x_{B_{i_2}} V factors through Y_i.
- Properties of diagonals of finite presentation. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-properties-diagonal and spaces-morphisms-lemma-diagonal-morphism-finite-type (cross-chapter, cited at ~5388-5391). E_i -> X_i is a finitely presented monomorphism.
- Descent of isomorphisms. lemma-descend-isomorphism (same chapter, cited at ~5399-5400). E_i = X_i for i large, hence f_i = g_i.

Source reference. Original equation 0CP8: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5264–5264. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CP4: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1024–1024. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0852: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1925–1925. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CP9: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5287–5287. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 04AK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 694–694. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CP6: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5249–5249. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084P: focused-seeds/geometry/S9-spaces-limits-morphism-good-diagram/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CP2: focused-seeds/geometry/S9-spaces-limits-morphism-good-diagram/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HK: focused-seeds/geometry/S9-spaces-limits-morphism-good-diagram/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0CP6. Let $f : X \to Y$ be a morphism of algebraic spaces quasi-separated and of finite type over B. Let*

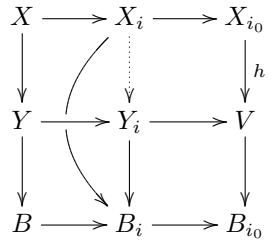
$$\begin{array}{ccc} X & \longrightarrow & W \\ \downarrow & & \downarrow \\ B & \longrightarrow & B_{i_1} \end{array} \quad \text{and} \quad \begin{array}{ccc} Y & \longrightarrow & V \\ \downarrow & & \downarrow \\ B & \longrightarrow & B_{i_2} \end{array}$$

be diagrams as in (0CP8). Let $X = \lim_{i \geq i_1} X_i$ and $Y = \lim_{i \geq i_2} Y_i$ be the corresponding limit descriptions as in Lemma 0CP9. Then there exists an $i_0 \geq \max(i_1, i_2)$ and a morphism

$$(f_i)_{i \geq i_0} : (X_i)_{i \geq i_0} \to (Y_i)_{i \geq i_0}$$

of inverse systems over $(B_i)_{i\geq i_0}$ such that such that $f = \lim_{i\geq i_0} f_i$. If $(g_i)_{i\geq i_0} : (X_i)_{i\geq i_0} \to (Y_i)_{i\geq i_0}$ is a second morphism of inverse systems over $(B_i)_{i\geq i_0}$ such that such that $f = \lim_{i\geq i_0} g_i$ then $f_i = g_i$ for all $i \gg i_0$.

Proof. Since $V \to B_{i_2}$ is of finite presentation and $X = \lim_{i\geq i_1} X_i$ we can appeal to Proposition 04AK as improved by Lemma 0CP4 to find an $i_0 \geq \max(i_1, i_2)$ and a morphism $h : X_{i_0} \to V$ over B_{i_2} such that $X \to X_{i_0} \to V$ is equal to $X \to Y \to V$. For $i \geq i_0$ we get a commutative solid diagram



Since $X \to X_i$ has scheme theoretically dense image and since Y_i is the scheme theoretic image of $Y \to B_i \times_{B_{i_2}} V$ we find that the morphism $X_i \to B_i \times_{B_{i_2}} V$ induced by the diagram factors through Y_i (Morphisms of Spaces, Lemma 0CP2). This proves existence.

Uniqueness. Let $E_i \to X_i$ be the equalizer of f_i and g_i for $i \geq i_0$. We have $E_i = Y_i \times_{\Delta, Y_i \times_{B_i} Y_i, (f_i, g_i)} X_i$. Hence $E_i \to X_i$ is a monomorphism of finite presentation as a base change of the diagonal of Y_i over B_i, see Morphisms of Spaces, Lemmas 03HK and 084P. Since X_i is a closed subspace of $B_i \times_{B_{i_0}} X_{i_0}$ and similarly for Y_i we see that

$$E_i = X_i \times_{(B_i \times_{B_{i_0}} X_{i_0})} (B_i \times_{B_{i_0}} E_{i_0}) = X_i \times_{X_{i_0}} E_{i_0}$$

Similarly, we have $X = X \times_{X_{i_0}} E_{i_0}$. Hence we conclude that $E_i = X_i$ for i large enough by Lemma 0852. □